

Ruby Setara

EDUCATION

Columbia University, Graduate School of Arts and Sciences, New York, NY

Master's of Arts, Biotechnology (GPA: 3.9/4.0)

September 2025 – October 2027

Fordham University, Fordham College, Bronx, NY

Bachelor of Science, Biological Sciences

September 2019 – May 2023

HONORS AND GRANTS

Society for Neuroscience Trainee Professional Development Award (2025)

Len Blavatnik STEM Research Fellowship (2023)

Fordham University Undergraduate Research Grant (Fall 2022)

Fordham University Undergraduate Research Grant (Summer 2022)

Jogues Scholarship (2019-2023)

RESEARCH EXPERIENCE

Columbia | Zuckerman Institute

Research Staff Assistant | Salzman Lab

May 2024 – Present

Advisors: Dr. Evan Kyzar (postdoctoral research scientist), Dr. Daniel Salzman (P.I.)

Conducting behavioral and histological experiments for several projects related to voluntary alcohol use, including disseminating the role of GLP1 receptor agonists in mitigating voluntary alcohol use and preference and defining the temporal progression of voluntary alcohol use in mice. Using immunohistochemistry, conditioned place preference and taste aversion assays, mouse handling and husbandry, chemogenetics, and confocal microscopy, we perform whole-brain cell-counting using an original pipeline to identify novel brain regions involved in alcohol consumption. Work presented at the Society for Neuroscience Conference 2024 & 2025.

Icahn School of Medicine at Mount Sinai

Research Associate | Becher Lab

July 2023 – January 2024

Advisors: Anna Racanelli, M.S., Dr. Oren Becher (P.I.)

Investigated novel therapeutic targets for diffuse intrinsic pontine glioma (DIPG), specifically the role of the RBPJ-SIX1 gene as a tumor suppressor. Using cell and neurosphere culture, we developed mouse models for DIPG, and conducted RNA extraction assays, Western blots, immunofluorescence, immunohistochemistry, and MRIs to characterize tumor growth. Mentored and trained high school interns on basic scientific research techniques. Work published in the Oxford Academic Neuro-Oncology Journal.

Fordham University, Department of Biological Sciences

Undergraduate Researcher | Neurobiology of Motivated Behavior Lab

September 2021 – May 2023

Advisor: Dr. Eduardo Gallo (P.I.)

Directed three independent research projects investigating the role of ventral pallidum cholinergic projection neurons (VP-CPNs) in cue-driven motivated behavior. Utilized in vivo fiber photometry to characterize VP-CPN activity during cue-motivated assays. Performed immunohistochemistry and data

analysis using GuPPy and MATLAB to identify VP-CPN activity on a trial-by-trial basis. Wrote and developed Fordham University Undergraduate Research Grant project proposals and operant behavioral assays. Mentored and trained new lab members ranging from technicians to undergraduate and graduate students. Work presented at the Society for Neuroscience Conference in Chicago, USA 2024.

Fordham University, Department of Psychology

Research Assistant

September 2020 – September 2021

Advisor: Dr. Andrew Rasmussen

Developed a multi-level Qualtrics survey assessing how sociodemographic and cultural factors in New York City college students predicted their perception of risk during the COVID-19 pandemic. Utilized SPSS and Excel to analyze and systematize 2,800+ response dataset. Developed and co-authored manuscript for publication.

Jacobi Medical Center | NYC Health + Hospitals

Research Assistant

January 2020 – December 2021

Advisor: Dr. Daniela Levanon

Coordinated the implementation of a weekly virtual lifestyle program during the COVID-19 pandemic targeting 126 low-income pediatric patients diagnosed with nonalcoholic fatty liver disease (NAFLD). Supported the medical staff by corresponding between patients and providers, conducted statistics on patient triglyceride levels, BMI, glycated hemoglobin, and other health markers using Excel and EPIC. Tracked patient biomarkers throughout the course of the program to characterize program efficacy.

LEADERSHIP AND SERVICE

Nucleate NY

Co-Director of Research

February 2026 – Present

Directing scientific and commercial due diligence on 100+ early-stage life science ventures for Nucleate NY's Activator program, evaluating technical viability, market fit, and clinical translational potential. Leveraging biotech deal analysis—including investment thesis evaluation and market sizing—to map the NYC venture capital ecosystem end-to-end and construct a strategic resource guide of 200+ target investors to accelerate seed-stage fundraising for accelerator cohorts. Co-authoring the annual Nucleate NY Impact Report, synthesizing quantitative programmatic data to communicate venture growth and ROI to external sponsors while benchmarking chapter performance to inform long-term programmatic strategy.

Biobus - Columbia University's Zuckerman Institute

Mentor

January 2025 – January 2026

Mentored high school students in researching for, developing, and presenting an independent research project as part of the Harlem Women In STEM Junior Scientist program. Students presented their work at the American Museum of Natural History in May 2025.

Columbia University Neuroscience Outreach (CUNO)

Saturday Science Planning Committee Member

September 2024 – September 2025

Collaborated with committee members to plan and execute themed neuroscience events for children in the local community. Developing child-friendly interactive educational activities aligning with a monthly theme.

PRESENTATIONS AND PUBLICATIONS

PUBLICATIONS

Gannon, A., **Setara, R.**, Maysonet, C. M., Salazar, R. E., Zuelke, D. R., Bowen, J. E., & Gallo, E. F. (2026). Ventral pallidum cholinergic neurons respond to reward and signal reward value. *bioRxiv*. <https://doi.org/10.64898/2026.05.24.726981>

Ramirez, A., Kyzar, E. J., Rogerson, L., Berland, C., Rodriguez, E., Guerrero, J., **Setara, R.**, Eisengart, M., Virkar, S., Hammond, L. A., Ferrante, A. W., & Salzman, C. D. (2026). A two-timepoint framework for sensitive and specific single-cell activity screening. *Neuron*, 114, 1–18. <https://doi.org/10.1016/j.neuron.2026.05.026>

Racanelli, A., Shen, C., Gadd, S. L., Chizhik, M., **Setara, R.**, Ge, Y., Schüpfer, A. J., Puigdelloses, M., Hambardzumyan, D., Tsankova, N. M., & Becher, O. J. (2024). DIPG-83. RBPJ-SIX1 as a tumor suppressor axis in diffuse intrinsic pontine glioma. *Neuro-Oncology*, 26(Supplement_4), iv38–iv39. <https://doi.org/10.1093/neuonc/noae064.136>

POSTERS

Setara, R., Kyzar, E., Eisengart, M., Virkar, S., Rogerson, L., & Salzman, C. D. (2025, November 15–19). *Brain-wide signatures of alcohol withdrawal using two-timepoint activity labeling*. Society for Neuroscience (SfN) 2025, San Diego, CA, United States.

E.J. Kyzar, **R. Setara**, L. Rogerson, M. Eisengart, A. Ramirez, B. Uceda-Alvarez, E. Rodriguez, C. Daniel Salzman; Whole-brain two-timepoint analysis of neuronal activity during the progression of voluntary alcohol drinking. LBA69. Oct.08.24. Chicago, Illinois, USA. Society for Neuroscience, 2024.

A. Gannon, **R. Setara**, R. Salazar, C. Maysonet, D. Zuelke, E.F. Gallo; Activity of ventral pallidum cholinergic projection neurons increases in response to reward and is modulated by reward value. PST296.25. Oct.08.24. Chicago, Illinois, USA. Society for Neuroscience, 2024.

Determining whether social cues are modulated by oxytocin receptor neurons. C. Cruz, E. Rodriguez, B. Uceda-Alvarez, **R. Setara**, M. Thurnherr, B. Jones and C. Daniel Salzman; New York, New York, USA. New York City Science Research and Mentoring Consortium, 2024.

Cue-evoked nucleus accumbens acetylcholine tracks reward value and delay cost in impulsive choice. J. Cavallaro, W. Mannheim, R. Salazar, **R. Setara**, C. Maysonet, E.F. Gallo; Ventura, California, USA. Basal Ganglia Gordon Research Conference, 2024.

Role of Ventral Pallidum Cholinergic Projection Neuron Activity in Cue-Driven Motivated Behavior. **R. Setara**, A. Gannon, R. Salazar, E.F. Gallo; Bronx, New York, USA. Fordham University Undergraduate Research Symposium, 2023.

J. Cavallaro, J. Yeisley, **R. Setara**, J. Floeder, E.F. Gallo; Biol. Sci., Fordham Univ., Bronx, NY.
Dopamine D2 receptors in nucleus accumbens cholinergic interneurons increase impulsive choice.
Program No. 649.06. 2022 Neuroscience Meeting Planner. San Diego, CA: Society for Neuroscience, 2022.

LABORATORY SKILLS

Animal behavior: Mouse handling and husbandry, operant conditioning, intraperitoneal and intracranial injections, fiber photometry, conditioned taste aversion, conditioned place preference, alcohol preference, in vivo optogenetics, Drosophila handling

Molecular: Immunohistochemistry, immunocytochemistry, Western blot, sectioning brain tissue on cryostat and vibratome, cell culture, RNA extraction, confocal microscopy, bioluminescent imaging, rodent MRIs

Computational: GraphPad Prism, ImageJ, BrainJ, Whole-Brain Reconstruction Pipeline, GuPPy, SPSS, MATLAB, EPIC, Qualtrics, spinning disk confocal microscopy, and Google Suite